

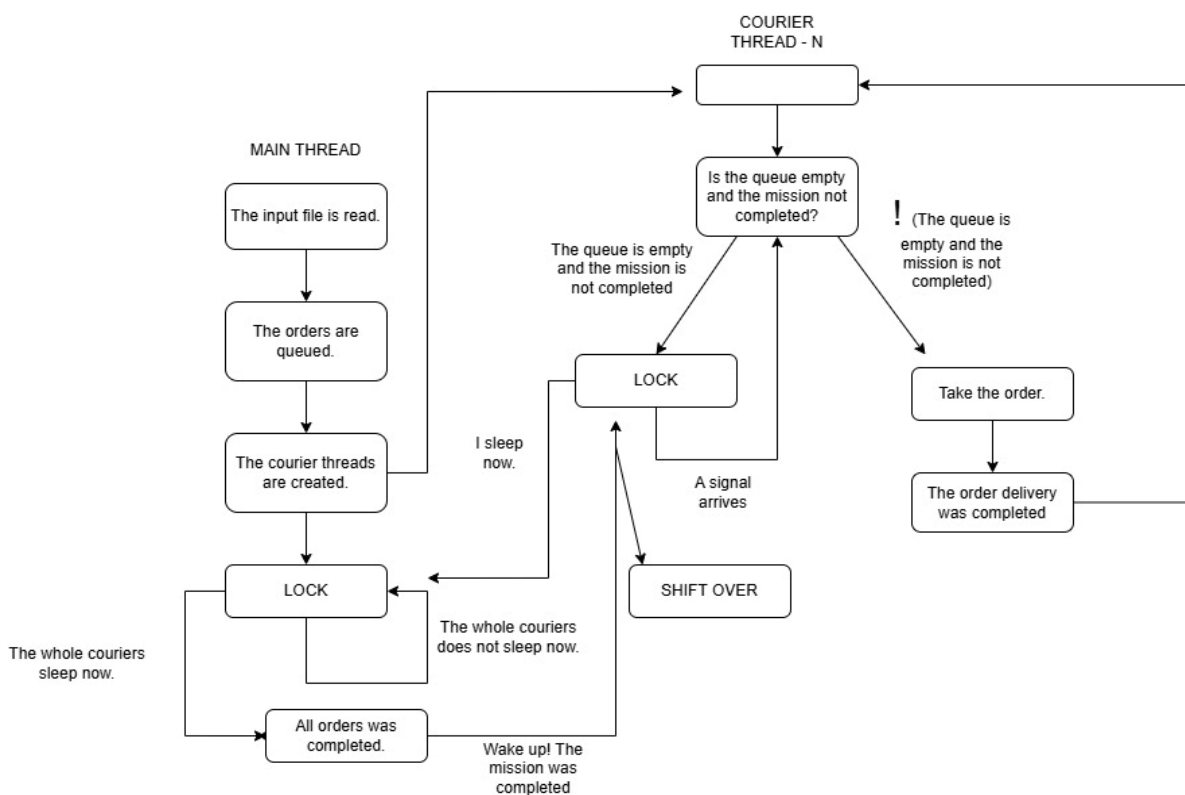
SYSTEM PROGRAMMING – HOMEWORK 5

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SYSTEM DESIGN



PRIORITY QUEUE DESIGN

The priority queue consists of order data structures. At every time the new order data is taken from the input file, the new element is appended to the end of the queue, and it is swapped with previous element constantly until its priority level is equal to or bigger than previous element's priority level.

When the priority queue is polled, just the front index is changed (front++) and the size is reduces (size--).

```

#ifndef ORDER_QUEUE_H
#define ORDER_QUEUE_H

#define MAX_NAME_LENGTH 32
#define MAX_LINE_LENGTH 70

typedef enum {
    EXPRESS, STANDARD, ECONOMY
} priority;

typedef struct {
    int id;
    char name[MAX_NAME_LENGTH];
    priority prt;
    int duration; // 1 unit = 500 milliseconds
} order;

typedef struct {
    order *orders;
    int size;
    int capacity;
    int front;
} order_queue; // priority queue

void read_order_list(order *item);
void initialize_order_list(int capacity);
void write_order_list(order *item);
void release_list();
void copy_order(order *to, order *from);
void swap_order(order *item_1, order *item_2);

#endif

```

SYNCHRONIZATION STRATEGY

In order to prevent race condition, some pthread_t mutexes and condition variables were used.

QUEUE MUTEX	Main thread, courier thread, signal thread
LOG MUTEX	Courier thread
NOT_EMPTY	Courier thread, main thread
FINISH	Main thread, courier thread, signal thread

The queue mutex is used when any order is read from the queue, the main thread and courier threads want to wake up each other, and signal thread send to finish signal to main thread.

Log mutex is benefited so that courier threads can write the events to the stdout.

not_empty condition variable is used for that when the courier threads finish their orders and they want to take new order, but the queue is empty at the same time, the couriers are put to sleep. The slept courier threads are woken up by main thread via not_empty cond. variable. (pthread_cond_broadcast(¬_empty)).

The threads about to sleep increment by 1 finish_counter and send finish signal to the main thread. The main thread taking the finish signal wake up and checks the condition (finish_counter == num_couriers). If any SIGINT signal, the signal handler thread increments the finish counter up to num_couriers illegally, and sends finish signal to the main thread.

SIGINT HANDLING

1. CTRL -C signal arrives.
2. The signal thread, which was asleep at sigwait(), wakes up.
3. The signal thread increases the finish counter (finish_counter = num_couriers).
4. It sends the finish signal to main thread.
5. The main thread wakes up, and checks the condition (finish_counter == num_couriers).
6. The main thread switches the is_completed flag to 1.

TEST SCENARIOS

1)

```
gcc -Wall -Wextra -std=c11 -pthread courier.o main.o order_queue.o -o program
muhammed@debian:~/Desktop/test/230104004017$ ./program -n 3 -i tests/10orders_mix.txt -s result.txt
[CARGOGTU] ORDER_QUEUED id = 1 recipient = Ahmet_Yilmaz priority =EXPRESS duration = 8
[CARGOGTU] ORDER_QUEUED id = 2 recipient = Ayse_Kaya priority =STANDARD duration = 12
[CARGOGTU] ORDER_QUEUED id = 3 recipient = Mehmet_Demir priority =ECONOMY duration = 20
[CARGOGTU] ORDER_QUEUED id = 4 recipient = Fatma_Sahin priority =EXPRESS duration = 6
[CARGOGTU] ORDER_QUEUED id = 5 recipient = Ali_Celik priority =STANDARD duration = 9
[CARGOGTU] ORDER_QUEUED id = 6 recipient = Zeynep_Arslan priority =ECONOMY duration = 15
[CARGOGTU] ORDER_QUEUED id = 7 recipient = Mustafa_Koc priority =EXPRESS duration = 7
[CARGOGTU] ORDER_QUEUED id = 8 recipient = Elif_Demir priority =STANDARD duration = 11
[CARGOGTU] ORDER_QUEUED id = 9 recipient = Hasan_Ozturk priority =ECONOMY duration = 18
[CARGOGTU] ORDER_QUEUED id = 10 recipient = Merve_Yildiz priority =STANDARD duration = 10
[CARGOGTU] SHIFT_START couriers = 3, orders = 10
[COURIER_1] DELIVERY_START id = 1 recipient = Ahmet_Yilmaz priority =EXPRESS
[COURIER_2] DELIVERY_START id = 4 recipient = Fatma_Sahin priority =EXPRESS
[COURIER_3] DELIVERY_START id = 7 recipient = Mustafa_Koc priority =EXPRESS
[COURIER_2] DELIVERY_COMPLETE id = 4 recipient = Fatma_Sahin duration = 3000ms
[COURIER_2] DELIVERY_START id = 2 recipient = Ayse_Kaya priority =STANDARD
[COURIER_3] DELIVERY_COMPLETE id = 7 recipient = Mustafa_Koc duration = 3500ms
[COURIER_3] DELIVERY_START id = 5 recipient = Ali_Celik priority =STANDARD
[COURIER_1] DELIVERY_COMPLETE id = 1 recipient = Ahmet_Yilmaz duration = 4000ms
[COURIER_1] DELIVERY_START id = 8 recipient = Elif_Demir priority =STANDARD
[COURIER_3] DELIVERY_COMPLETE id = 5 recipient = Ali_Celik duration = 4500ms
[COURIER_3] DELIVERY_START id = 10 recipient = Merve_Yildiz priority =STANDARD
[COURIER_2] DELIVERY_COMPLETE id = 2 recipient = Ayse_Kaya duration = 6000ms
[COURIER_2] DELIVERY_START id = 3 recipient = Mehmet_Demir priority =ECONOMY
[COURIER_1] DELIVERY_COMPLETE id = 8 recipient = Elif_Demir duration = 5500ms
[COURIER_1] DELIVERY_START id = 6 recipient = Zeynep_Arslan priority =ECONOMY
[COURIER_3] DELIVERY_COMPLETE id = 10 recipient = Merve_Yildiz duration = 5000ms
[COURIER_3] DELIVERY_START id = 9 recipient = Hasan_Ozturk priority =ECONOMY
[COURIER_1] DELIVERY_COMPLETE id = 6 recipient = Zeynep_Arslan duration = 7500ms
[COURIER_1] WAITING
[COURIER_2] DELIVERY_COMPLETE id = 3 recipient = Mehmet_Demir duration = 10000ms
[COURIER_2] WAITING
[COURIER_3] DELIVERY_COMPLETE id = 9 recipient = Hasan_Ozturk duration = 9000ms
[COURIER_3] WAITING
[COURIER_2] SHIFT_OVER
[COURIER_3] SHIFT_OVER
[COURIER_1] SHIFT_OVER
[CARGOGTU] SHIFT_END completed = 10 cancelled = 0 total_time = 58000
[CARGOGTU] SHUTDOWN_COMPLETE
muhammed@debian:~/Desktop/test/230104004017$
```

```

Activities Text Editor
Open [icon]
1 SHIFT_SUMMARY
2 Total orders : 10
3 Completed : 10
4 Cancelled : 0
5 Total time : 58000ms
6 Avg per order : 5800ms
7
8 COURIER_STATS
9 Courier-1 completed = 3 total_time = 17000ms
10 Courier-2 completed = 3 total_time = 19000ms
11 Courier-3 completed = 4 total_time = 22000ms

```

2)

```

muhammed@debian:~/Desktop/test/230104004017$ ./program -n 10 -i tests/10orders_economy.txt -s result.txt
[CARGOGTU] ORDER QUEUED id = 1 recipient = Ahmet Yilmaz priority =ECONOMY duration = 8
[CARGOGTU] ORDER QUEUED id = 2 recipient = Ayse Kaya priority =ECONOMY duration = 10
[CARGOGTU] ORDER QUEUED id = 3 recipient = Mehmet Demir priority =ECONOMY duration = 7
[CARGOGTU] ORDER QUEUED id = 4 recipient = Fatma Sahin priority =ECONOMY duration = 12
[CARGOGTU] ORDER QUEUED id = 5 recipient = Ali Celik priority =ECONOMY duration = 9
[CARGOGTU] ORDER QUEUED id = 6 recipient = Zeynep Arslan priority =ECONOMY duration = 11
[CARGOGTU] ORDER QUEUED id = 7 recipient = Mustafa Koc priority =ECONOMY duration = 8
[CARGOGTU] ORDER QUEUED id = 8 recipient = Elif Demir priority =ECONOMY duration = 6
[CARGOGTU] ORDER QUEUED id = 9 recipient = Hasan Ozturk priority =ECONOMY duration = 10
[CARGOGTU] ORDER QUEUED id = 10 recipient = Merve Yildiz priority =ECONOMY duration = 9
[CARGOGTU] SHIFT START couriers = 10, orders = 10
[COURIER 1] DELIVERY_START id = 1 recipient = Ahmet Yilmaz priority =ECONOMY
[COURIER 2] DELIVERY_START id = 2 recipient = Ayse Kaya priority =ECONOMY
[COURIER 5] DELIVERY_START id = 3 recipient = Mehmet Demir priority =ECONOMY
[COURIER 3] DELIVERY_START id = 4 recipient = Fatma Sahin priority =ECONOMY
[COURIER 6] DELIVERY_START id = 5 recipient = Ali Celik priority =ECONOMY
[COURIER 9] DELIVERY_START id = 8 recipient = Elif Demir priority =ECONOMY
[COURIER 8] DELIVERY_START id = 9 recipient = Hasan Ozturk priority =ECONOMY
[COURIER 7] DELIVERY_START id = 6 recipient = Zeynep Arslan priority =ECONOMY
[COURIER 4] DELIVERY_START id = 7 recipient = Mustafa Koc priority =ECONOMY
[COURIER 10] DELIVERY_START id = 10 recipient = Merve Yildiz priority =ECONOMY
[COURIER 9] DELIVERY_COMPLETE id = 8 recipient = Elif Demir duration = 3000ms
[COURIER-9] WAITING
[COURIER 5] DELIVERY_COMPLETE id = 3 recipient = Mehmet Demir duration = 3500ms
[COURIER-5] WAITING
[COURIER 1] DELIVERY_COMPLETE id = 1 recipient = Ahmet Yilmaz duration = 4000ms
[COURIER-1] WAITING
[COURIER 4] DELIVERY_COMPLETE id = 7 recipient = Mustafa Koc duration = 4000ms
[COURIER-4] WAITING
[COURIER 6] DELIVERY_COMPLETE id = 5 recipient = Ali Celik duration = 4500ms
[COURIER-6] WAITING
[COURIER 10] DELIVERY_COMPLETE id = 10 recipient = Merve Yildiz duration = 4500ms
[COURIER-10] WAITING
[COURIER 2] DELIVERY_COMPLETE id = 2 recipient = Ayse Kaya duration = 5000ms
[COURIER-2] WAITING
[COURIER 8] DELIVERY_COMPLETE id = 9 recipient = Hasan Ozturk duration = 5000ms
[COURIER-8] WAITING
[COURIER 7] DELIVERY_COMPLETE id = 6 recipient = Zeynep Arslan duration = 5500ms
[COURIER-7] WAITING
[COURIER 3] DELIVERY_COMPLETE id = 4 recipient = Fatma Sahin duration = 6000ms
[COURIER-3] WAITING
[COURIER-2] SHIFT_OVER
[COURIER-4] SHIFT_OVER
[COURIER-1] SHIFT_OVER
[COURIER-3] SHIFT_OVER
[COURIER-9] SHIFT_OVER
[COURIER-6] SHIFT_OVER
[COURIER-8] SHIFT_OVER
[COURIER-5] SHIFT_OVER
[COURIER-7] SHIFT_OVER
[COURIER-10] SHIFT_OVER
[CARGOGTU] SHIFT_END completed = 10 cancelled = 0 total_time = 45000
[CARGOGTU] SHUTDOWN COMPLETE
muhammed@debian:~/Desktop/test/230104004017$ ./program -n 2 -i tests/20orders_mix.txt -s result.txt
[CARGOGTU] ORDER QUEUED id = 1 recipient = Ahmet Yilmaz priority =ECONOMY duration = 1

```

```

Open ▼
1 SHIFT_SUMMARY
2 Total orders : 10
3 Completed : 10
4 Cancelled : 0
5 Total time : 45000ms
6 Avg per order : 4500ms
7
8 COURIER_STATS
9 Courier-1 completed = 1 total_time = 4000ms
10 Courier-2 completed = 1 total_time = 5000ms
11 Courier-3 completed = 1 total_time = 3500ms
12 Courier-4 completed = 1 total_time = 5500ms
13 Courier-5 completed = 1 total_time = 4500ms
14 Courier-6 completed = 1 total_time = 6000ms
15 Courier-7 completed = 1 total_time = 4000ms
16 Courier-8 completed = 1 total_time = 3000ms
17 Courier-9 completed = 1 total_time = 4500ms
18 Courier-10 completed = 1 total_time = 5000ms

```

3)

```

nuhammed@debian:~/Desktop/test/230104004017$ ./program -n 2 -i tests/20orders_mix.txt -s result.txt
[CARGOGTU] ORDER_QUEUED id = 1 recipient = Ahmet_Yilmaz priority =EXPRESS duration = 1
[CARGOGTU] ORDER_QUEUED id = 2 recipient = Ayse_Kaya priority =ECONOMY duration = 3
[CARGOGTU] ORDER_QUEUED id = 3 recipient = Mehmet_Demir priority =STANDARD duration = 2
[CARGOGTU] ORDER_QUEUED id = 4 recipient = Fatma_Sahin priority =EXPRESS duration = 1
[CARGOGTU] ORDER_QUEUED id = 5 recipient = Ali_Celik priority =STANDARD duration = 3
[CARGOGTU] ORDER_QUEUED id = 6 recipient = Zeynep_Arslan priority =ECONOMY duration = 2
[CARGOGTU] ORDER_QUEUED id = 7 recipient = Mustafa_Koc priority =EXPRESS duration = 1
[CARGOGTU] ORDER_QUEUED id = 8 recipient = Elif_Demir priority =STANDARD duration = 2
[CARGOGTU] ORDER_QUEUED id = 9 recipient = Hasan_Ozturk priority =ECONOMY duration = 3
[CARGOGTU] ORDER_QUEUED id = 10 recipient = Merve_Yildiz priority =EXPRESS duration = 1
[CARGOGTU] ORDER_QUEUED id = 11 recipient = Emre_Celik priority =STANDARD duration = 2
[CARGOGTU] ORDER_QUEUED id = 12 recipient = Selin_Kaya priority =ECONOMY duration = 3
[CARGOGTU] ORDER_QUEUED id = 13 recipient = Burak_Arslan priority =EXPRESS duration = 1
[CARGOGTU] ORDER_QUEUED id = 14 recipient = Dilan_Ozkan priority =STANDARD duration = 2
[CARGOGTU] ORDER_QUEUED id = 15 recipient = Kemal_Sahin priority =ECONOMY duration = 3
[CARGOGTU] ORDER_QUEUED id = 16 recipient = Rana_Yilmaz priority =EXPRESS duration = 2
[CARGOGTU] ORDER_QUEUED id = 17 recipient = Tarik_Demir priority =STANDARD duration = 1
[CARGOGTU] ORDER_QUEUED id = 18 recipient = Pinar_Koc priority =ECONOMY duration = 2
[CARGOGTU] ORDER_QUEUED id = 19 recipient = Oguz_Yildiz priority =EXPRESS duration = 3
[CARGOGTU] ORDER_QUEUED id = 20 recipient = Ceren_Arslan priority =STANDARD duration = 1
[CARGOGTU] SHIFT_START couriers = 2, orders = 20
[COURIER_1] DELIVERY_START id = 1 recipient = Ahmet_Yilmaz priority =EXPRESS
[COURIER_2] DELIVERY_START id = 4 recipient = Fatma_Sahin priority =EXPRESS
[COURIER_1] DELIVERY_COMPLETE id = 1 recipient = Ahmet_Yilmaz duration = 500ms
[COURIER_1] DELIVERY_START id = 7 recipient = Mustafa_Koc priority =EXPRESS
[COURIER_2] DELIVERY_COMPLETE id = 4 recipient = Fatma_Sahin duration = 500ms
[COURIER_2] DELIVERY_START id = 10 recipient = Merve_Yildiz priority =EXPRESS
[COURIER_1] DELIVERY_COMPLETE id = 7 recipient = Mustafa_Koc duration = 500ms
[COURIER_1] DELIVERY_START id = 13 recipient = Burak_Arslan priority =EXPRESS
[COURIER_2] DELIVERY_COMPLETE id = 10 recipient = Merve_Yildiz duration = 500ms
[COURIER_2] DELIVERY_START id = 16 recipient = Rana_Yilmaz priority =EXPRESS
[COURIER_1] DELIVERY_COMPLETE id = 13 recipient = Burak_Arslan duration = 500ms
[COURIER_1] DELIVERY_START id = 19 recipient = Oguz_Yildiz priority =EXPRESS
^C[COURIER_2] DELIVERY_COMPLETE id = 16 recipient = Rana_Yilmaz duration = 1000ms
[COURIER-2] SHIFT_OVER
[COURIER_1] DELIVERY_COMPLETE id = 19 recipient = Oguz_Yildiz duration = 1500ms
[COURIER-1] SHIFT_OVER
[CARGOGTU] SHIFT_END completed = 7 cancelled = 13 total_time = 5000
[CARGOGTU] SHUTDOWN_COMPLETE
nuhammed@debian:~/Desktop/test/230104004017$ ps
  PID TTY          TIME CMD
 23424 pts/0    00:00:00 bash
 25035 pts/0    00:00:00 ps
nuhammed@debian:~/Desktop/test/230104004017$

```



```

1 SHIFT_SUMMARY
2 Total orders : 20
3 Completed : 7
4 Cancelled : 13
5 Total time : 5000ms
6 Avg per order : 714ms
7
8 COURIER STATS
9 Courier-1 completed = 4 total_time = 3000ms
10 Courier-2 completed = 3 total_time = 2000ms

```

4)

```

==25883== ERROR SUMMARY: 1 errors from 1 contexts (suppressed: 0 from 0)
muhammed@debian:~/Desktop/test/230104004017$ valgrind --leak-check=full ./program -n 3 -i tests/10orders_mix.txt -s result.txt
==25883== Memcheck, a memory error detector
==25883== Copyright (C) 2002-2017, and GNU GPL'd, by Julian Seward et al.
==25883== Using Valgrind-3.16.1 and LibVEX; rerun with -h for copyright info
==25883== Command: ./program -n 3 -i tests/10orders_mix.txt -s result.txt
==25883==
[CARGOGTU] ORDER_QUEUED id = 1 recipient = Ahmet Yilmaz priority =EXPRESS duration = 8
[CARGOGTU] ORDER_QUEUED id = 2 recipient = Ayse Kaya priority =STANDARD duration = 12
[CARGOGTU] ORDER_QUEUED id = 3 recipient = Mehmet Demir priority =ECONOMY duration = 20
[CARGOGTU] ORDER_QUEUED id = 4 recipient = Fatma Sahin priority =EXPRESS duration = 6
[CARGOGTU] ORDER_QUEUED id = 5 recipient = Ali Celik priority =STANDARD duration = 9
[CARGOGTU] ORDER_QUEUED id = 6 recipient = Zeynep Arslan priority =ECONOMY duration = 15
[CARGOGTU] ORDER_QUEUED id = 7 recipient = Mustafa Koc priority =EXPRESS duration = 7
[CARGOGTU] ORDER_QUEUED id = 8 recipient = Elif Demir priority =STANDARD duration = 11
[CARGOGTU] ORDER_QUEUED id = 9 recipient = Hasan Ozturk priority =ECONOMY duration = 18
[CARGOGTU] ORDER_QUEUED id = 10 recipient = Merve Yildiz priority =STANDARD duration = 10
[CARGOGTU] SHIFT_START couriers = 3, orders = 10
[COURIER_3] DELIVERY_START id = 1 recipient = Ahmet Yilmaz priority =EXPRESS
[COURIER_1] DELIVERY_START id = 4 recipient = Fatma Sahin priority =EXPRESS
[COURIER_2] DELIVERY_START id = 7 recipient = Mustafa Koc priority =EXPRESS
[COURIER_1] DELIVERY_COMPLETE id = 4 recipient = Fatma Sahin duration = 3000ms
[COURIER_1] DELIVERY_START id = 2 recipient = Ayse Kaya priority =STANDARD
[COURIER_2] DELIVERY_COMPLETE id = 7 recipient = Mustafa Koc duration = 3500ms
[COURIER_2] DELIVERY_START id = 5 recipient = Ali Celik priority =STANDARD
[COURIER_3] DELIVERY_COMPLETE id = 1 recipient = Ahmet Yilmaz duration = 4000ms
[COURIER_3] DELIVERY_START id = 8 recipient = Elif Demir priority =STANDARD
[COURIER_2] DELIVERY_COMPLETE id = 5 recipient = Ali Celik duration = 4500ms
[COURIER_2] DELIVERY_START id = 10 recipient = Merve Yildiz priority =STANDARD
[COURIER_1] DELIVERY_COMPLETE id = 2 recipient = Ayse Kaya duration = 6000ms
[COURIER_1] DELIVERY_START id = 3 recipient = Mehmet Demir priority =ECONOMY
[COURIER_3] DELIVERY_COMPLETE id = 8 recipient = Elif Demir duration = 5500ms
[COURIER_3] DELIVERY_START id = 6 recipient = Zeynep Arslan priority =ECONOMY
[COURIER_2] DELIVERY_COMPLETE id = 10 recipient = Merve Yildiz duration = 5000ms
[COURIER_2] DELIVERY_START id = 9 recipient = Hasan Ozturk priority =ECONOMY
[COURIER_3] DELIVERY_COMPLETE id = 6 recipient = Zeynep Arslan duration = 7500ms
[COURIER_3] WAITING
[COURIER_1] DELIVERY_COMPLETE id = 3 recipient = Mehmet Demir duration = 10000ms
[COURIER_1] WAITING
[COURIER_2] DELIVERY_COMPLETE id = 9 recipient = Hasan Ozturk duration = 9000ms
[COURIER_2] WAITING
[COURIER_3] SHIFT_OVER
[COURIER_1] SHIFT_OVER
[COURIER_2] SHIFT_OVER
[CARGOGTU] SHIFT_END completed = 10 cancelled = 0 total_time = 58000
[CARGOGTU] SHUTDOWN_COMPLETE
==25883==
==25883== HEAP SUMMARY:
==25883==   in use at exit: 272 bytes in 1 blocks
==25883==   total heap usage: 13 allocs, 12 frees, 11,772 bytes allocated
==25883==
==25883== 272 bytes in 1 blocks are possibly lost in loss record 1 of 1
==25883==   at 0x483AB65: calloc (vg_replace_malloc.c:760)
==25883==   by 0x4012D16: allocate_dtv (dl-tls.c:343)
==25883==   by 0x4012D16: _dl_allocate_tls (dl-tls.c:589)
==25883==   by 0x4864B81: allocate_stack (allocatestack.c:622)
==25883==   by 0x4864B81: pthread_create@@GLIBC_2.2.5 (pthread_create.c:660)
==25883==   by 0x109799: courier_main (in /home/muhammed/Desktop/test/230104004017/program)
==25883==   by 0x10A0AE: main (in /home/muhammed/Desktop/test/230104004017/program)
==25883==
==25883== LEAK SUMMARY:
==25883==   definitely lost: 0 bytes in 0 blocks
==25883==   indirectly lost: 0 bytes in 0 blocks
==25883==   possibly lost: 272 bytes in 1 blocks
==25883==   still reachable: 0 bytes in 0 blocks
==25883==   suppressed: 0 bytes in 0 blocks
==25883==
==25883== For lists of detected and suppressed errors, rerun with: -s
==25883== ERROR SUMMARY: 1 errors from 1 contexts (suppressed: 0 from 0)
muhammed@debian:~/Desktop/test/230104004017$

```



```

[CARGOGTU] SHUTDOWN_COMPLETE
muhammed@debian:~/Desktop/test/230104004017$ ./program -n 10 -i tests/20orders_mix.txt -s result.txt
[CARGOGTU] ORDER_QUEUED id = 1 recipient = Ahmet_Yilmaz priority =EXPRESS duration = 1
[CARGOGTU] ORDER_QUEUED id = 2 recipient = Ayse_Kaya priority =ECONOMY duration = 3
[CARGOGTU] ORDER_QUEUED id = 3 recipient = Mehmet_Demir priority =STANDARD duration = 2
[CARGOGTU] ORDER_QUEUED id = 4 recipient = Fatma_Sahin priority =EXPRESS duration = 1
[CARGOGTU] ORDER_QUEUED id = 5 recipient = Ali_Celik priority =STANDARD duration = 3
[CARGOGTU] ORDER_QUEUED id = 6 recipient = Zeynep_Arslan priority =ECONOMY duration = 2
[CARGOGTU] ORDER_QUEUED id = 7 recipient = Mustafa_Koc priority =EXPRESS duration = 1
[CARGOGTU] ORDER_QUEUED id = 8 recipient = Elif_Demir priority =STANDARD duration = 2
[CARGOGTU] ORDER_QUEUED id = 9 recipient = Hasan_Ozturk priority =ECONOMY duration = 3
[CARGOGTU] ORDER_QUEUED id = 10 recipient = Merve_Yildiz priority =EXPRESS duration = 1
[CARGOGTU] ORDER_QUEUED id = 11 recipient = Enre_Celik priority =STANDARD duration = 2
[CARGOGTU] ORDER_QUEUED id = 12 recipient = Selin_Kaya priority =ECONOMY duration = 3
[CARGOGTU] ORDER_QUEUED id = 13 recipient = Burak_Arslan priority =EXPRESS duration = 1
[CARGOGTU] ORDER_QUEUED id = 14 recipient = Dilan_Ozkan priority =STANDARD duration = 2
[CARGOGTU] ORDER_QUEUED id = 15 recipient = Kemal_Sahin priority =ECONOMY duration = 3
[CARGOGTU] ORDER_QUEUED id = 16 recipient = Rana_Yilmaz priority =EXPRESS duration = 2
[CARGOGTU] ORDER_QUEUED id = 17 recipient = Tarik_Demir priority =STANDARD duration = 1
[CARGOGTU] ORDER_QUEUED id = 18 recipient = Pinar_Koc priority =ECONOMY duration = 2
[CARGOGTU] ORDER_QUEUED id = 19 recipient = Oguz_Yildiz priority =EXPRESS duration = 3
[CARGOGTU] ORDER_QUEUED id = 20 recipient = Ceren_Arslan priority =STANDARD duration = 1
[CARGOGTU] SHIFT_START couriers = 10, orders = 20
[COURIER_1] DELIVERY_START id = 1 recipient = Ahmet_Yilmaz priority =EXPRESS
[COURIER_4] DELIVERY_START id = 4 recipient = Fatma_Sahin priority =EXPRESS
[COURIER_6] DELIVERY_START id = 7 recipient = Mustafa_Koc priority =EXPRESS
[COURIER_5] DELIVERY_START id = 10 recipient = Merve_Yildiz priority =EXPRESS
[COURIER_7] DELIVERY_START id = 13 recipient = Burak_Arslan priority =EXPRESS
[COURIER_9] DELIVERY_START id = 16 recipient = Rana_Yilmaz priority =EXPRESS
[COURIER_8] DELIVERY_START id = 19 recipient = Oguz_Yildiz priority =EXPRESS
[COURIER_2] DELIVERY_START id = 3 recipient = Mehmet_Demir priority =STANDARD
[COURIER_10] DELIVERY_START id = 8 recipient = Elif_Demir priority =STANDARD
[COURIER_3] DELIVERY_START id = 5 recipient = Ali_Celik priority =STANDARD
[COURIER_1] DELIVERY_COMPLETE id = 1 recipient = Ahmet_Yilmaz duration = 500ms
[COURIER_1] DELIVERY_START id = 11 recipient = Enre_Celik priority =STANDARD
[COURIER_6] DELIVERY_COMPLETE id = 7 recipient = Mustafa_Koc duration = 500ms
[COURIER_6] DELIVERY_START id = 14 recipient = Dilan_Ozkan priority =STANDARD
[COURIER_4] DELIVERY_COMPLETE id = 4 recipient = Fatma_Sahin duration = 500ms
[COURIER_4] DELIVERY_START id = 17 recipient = Tarik_Demir priority =STANDARD
[COURIER_7] DELIVERY_COMPLETE id = 13 recipient = Burak_Arslan duration = 500ms
[COURIER_7] DELIVERY_START id = 20 recipient = Ceren_Arslan priority =STANDARD
[COURIER_5] DELIVERY_COMPLETE id = 10 recipient = Merve_Yildiz duration = 500ms
[COURIER_5] DELIVERY_START id = 2 recipient = Ayse_Kaya priority =ECONOMY
[COURIER_9] DELIVERY_COMPLETE id = 16 recipient = Rana_Yilmaz duration = 1000ms
[COURIER_9] DELIVERY_START id = 6 recipient = Zeynep_Arslan priority =ECONOMY
[COURIER_2] DELIVERY_COMPLETE id = 3 recipient = Mehmet_Demir duration = 1000ms
[COURIER_2] DELIVERY_START id = 9 recipient = Hasan_Ozturk priority =ECONOMY
[COURIER_10] DELIVERY_COMPLETE id = 8 recipient = Elif_Demir duration = 1000ms
[COURIER_10] DELIVERY_START id = 12 recipient = Selin_Kaya priority =ECONOMY
[COURIER_4] DELIVERY_COMPLETE id = 17 recipient = Tarik_Demir duration = 500ms
[COURIER_4] DELIVERY_START id = 15 recipient = Kemal_Sahin priority =ECONOMY
[COURIER_7] DELIVERY_COMPLETE id = 20 recipient = Ceren_Arslan duration = 500ms
[COURIER_7] DELIVERY_START id = 18 recipient = Pinar_Koc priority =ECONOMY
[COURIER_8] DELIVERY_COMPLETE id = 19 recipient = Oguz_Yildiz duration = 1500ms
[COURIER_8] WAITING
[COURIER_3] DELIVERY_COMPLETE id = 5 recipient = Ali_Celik duration = 1500ms
[COURIER_3] WAITING
[COURIER_1] DELIVERY_COMPLETE id = 11 recipient = Enre_Celik duration = 1000ms
[COURIER_1] WAITING
[COURIER_6] DELIVERY_COMPLETE id = 14 recipient = Dilan_Ozkan duration = 1000ms
[COURIER_6] WAITING
[COURIER_5] DELIVERY_COMPLETE id = 2 recipient = Ayse_Kaya duration = 1500ms
[COURIER_5] WAITING
[COURIER_9] DELIVERY_COMPLETE id = 6 recipient = Zeynep_Arslan duration = 1000ms
[COURIER_9] WAITING
[COURIER_7] DELIVERY_COMPLETE id = 18 recipient = Pinar_Koc duration = 1000ms
[COURIER_7] WAITING
[COURIER_2] DELIVERY_COMPLETE id = 9 recipient = Hasan_Ozturk duration = 1500ms
[COURIER_2] WAITING
[COURIER_10] DELIVERY_COMPLETE id = 12 recipient = Selin_Kaya duration = 1500ms
[COURIER_10] WAITING
[COURIER_4] DELIVERY_COMPLETE id = 15 recipient = Kemal_Sahin duration = 1500ms
[COURIER_4] WAITING
[COURIER_8] SHIFT_OVER
[COURIER_6] SHIFT_OVER
[COURIER_9] SHIFT_OVER
[COURIER_2] SHIFT_OVER
[COURIER_4] SHIFT_OVER
[COURIER_1] SHIFT_OVER
[COURIER_10] SHIFT_OVER
[COURIER_3] SHIFT_OVER
[COURIER_7] SHIFT_OVER
[COURIER_5] SHIFT_OVER
[CARGOGTU] SHIFT_END completed = 20 cancelled = 0 total_time = 19500
[CARGOGTU] SHUTDOWN_COMPLETE
muhammed@debian:~/Desktop/test/230104004017$

```


CHALLENGES AND SOLUTIONS

1) Delivery Time Problem:

Because of delivery of order time, the program would need to proceed throughout appreciable time. However, the program was being completed very fast. The solution of this problem is very easy:

```
rq.tv_sec = item.duration * 500 / 1000;
rq.tv_nsec = (item.duration * 500 - rq.tv_sec * pow(10, 3)) * pow(10, 6);

nanosleep(&rq, &rm); // delivery operation
```

It was just arithmetic mistake.

2) Is the order of the queue correct?

It can be seen that the order of delivery is incorrect at some running. Alright, is it expected, or unexpected? In multithreading programs like this project, the behaviour of flow of control is non-deterministic, in other words, a thread takes the order, another thread takes following order, prints the state and the first thread prints own state, that is possible.

A program shows a test about order queue and its output:

```
#include "order_queue.h"
#include <stdio.h>
#include <unistd.h>
#include <string.h>
#include "courier.h"

int main() {
    order item[20];
    order item_1;

    initialize_order_list(20);

    for(int i = 0; i < 20; i++) {
        item[i].prt = (i % 3 == 0) ? ECONOMY : (i % 3 == 1) ? EXPRESS : (i % 3 == 2) ? STANDARD : ECONOMY;
        item[i].id = i + 1;
        sprintf(item[i].name, "test %d", i + 1);
        item[i].duration = 10;
        printf("id: %d name %s priority:", item[i].id, item[i].name);
        print_priority(item[i].prt);
        printf("\n");
        write_order_list(&item[i]);
    }

    printf("\n");

    for(int i = 0; i < 20; i++) {
        read_order_list(&item_1);
        printf("id: %d name %s priority:", item_1.id, item_1.name);
        print_priority(item_1.prt);
        printf("\n");
    }

    release_list();

    return 0;
}
```

```
muhammed_baslak@BASLAK:~/programming/system/HW5/230104004017$ ./test_prog
id: 3 name test 3 priority:STANDARD
id: 4 name test 4 priority:ECONOMY
id: 5 name test 5 priority:EXPRESS
id: 6 name test 6 priority:STANDARD
id: 7 name test 7 priority:ECONOMY
id: 8 name test 8 priority:EXPRESS
id: 9 name test 9 priority:STANDARD
id: 10 name test 10 priority:ECONOMY
id: 11 name test 11 priority:EXPRESS
id: 12 name test 12 priority:STANDARD
id: 13 name test 13 priority:ECONOMY
id: 14 name test 14 priority:EXPRESS
id: 15 name test 15 priority:STANDARD
id: 16 name test 16 priority:ECONOMY
id: 17 name test 17 priority:EXPRESS
id: 18 name test 18 priority:STANDARD
id: 19 name test 19 priority:ECONOMY
id: 20 name test 20 priority:EXPRESS

id: 2 name test 2 priority:EXPRESS
id: 5 name test 5 priority:EXPRESS
id: 8 name test 8 priority:EXPRESS
id: 11 name test 11 priority:EXPRESS
id: 14 name test 14 priority:EXPRESS
id: 17 name test 17 priority:EXPRESS
id: 20 name test 20 priority:EXPRESS
id: 3 name test 3 priority:STANDARD
id: 6 name test 6 priority:STANDARD
id: 9 name test 9 priority:STANDARD
id: 12 name test 12 priority:STANDARD
id: 15 name test 15 priority:STANDARD
id: 18 name test 18 priority:STANDARD
id: 1 name test 1 priority:ECONOMY
id: 4 name test 4 priority:ECONOMY
id: 7 name test 7 priority:ECONOMY
id: 10 name test 10 priority:ECONOMY
id: 13 name test 13 priority:ECONOMY
id: 16 name test 16 priority:ECONOMY
id: 19 name test 19 priority:ECONOMY
muhammed_baslak@BASLAK:~/programming/system/HW5/230104004017$
```